Lab Experiment-19

Aim

To write a Python program to perform Word Sense Disambiguation (WSD) using the Lesk algorithm to identify the correct meaning of an ambiguous word based on its context.

Algorithm

1. Import the required libraries from NLTK.
2. Download the WordNet dataset.
3. Get a sentence and an ambiguous word as input from the user.
4. Retrieve all possible meanings (synsets) of the word from WordNet.
5. Compare the sentence context with the definitions and examples of each meaning.
6. Calculate the overlap between the context words and each meaning.
7. Select the meaning with the highest overlap.
8. Display the most appropriate meaning.

Code:

```
import nltk

from nltk.wsd import lesk

from nltk.corpus import wordnet

nltk.download('wordnet')

nltk.download('omw-1.4')

sentence = input("Enter a sentence: ")

word = input("Enter the ambiguous word: ")
```

```
tokens = nltk.word_tokenize(sentence)

sense = lesk(tokens, word)

if sense:

    print("\nWord:", word)

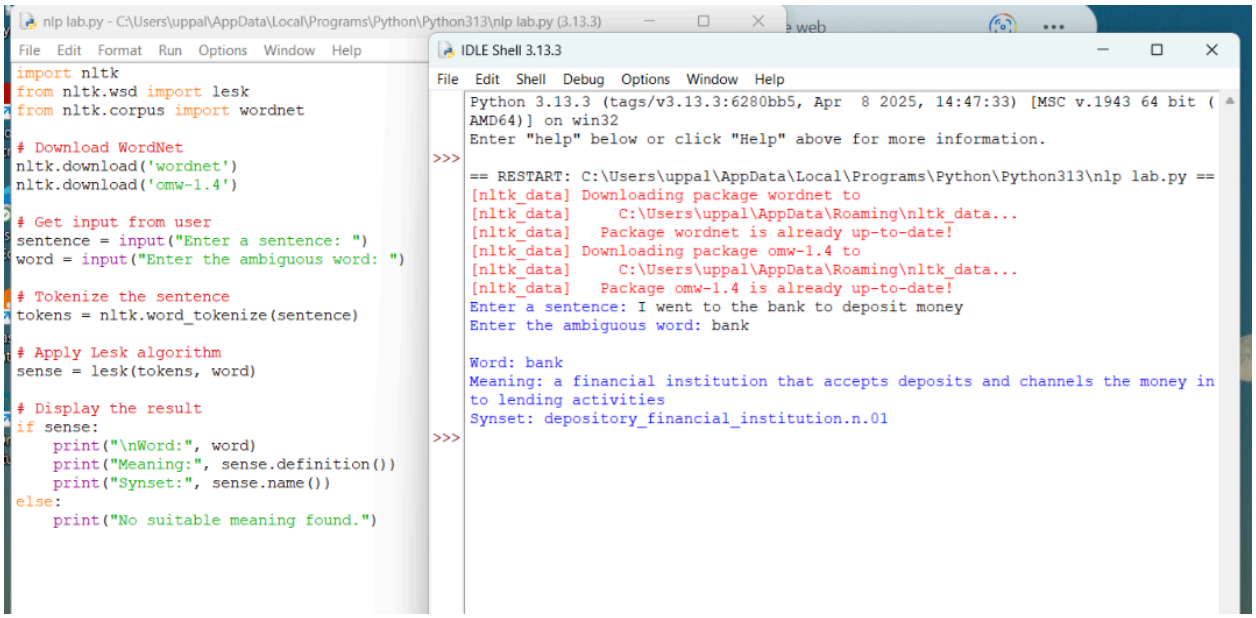
    print("Meaning:", sense.definition())

    print("Synset:", sense.name())

else:

    print("No suitable meaning found.")
```

Output:



Result

Thus, the Python program successfully performs Word Sense Disambiguation using the Lesk algorithm and identifies the appropriate meaning of an ambiguous word based on its context.